

DAILY REPORT OF INTERNSHIP AT IIIT NAYA RAIPUR OUTREACH PROGRAM 45 DAYS

REGISTRATION ID: 20260027

DAY-01 [27052026]

Objective:

IMPLEMENTATION OF TRANSFER AND DRAIN CHARACTERISTICS OF NMOS AND PMOS IN PSPICE

Theoretical Background:

MOSFET (Metal Oxide Semiconductor Field Effect Transistor)

Definition: A MOSFET works as an electronic switching device that controls the flow of current in a circuit by the gate voltage (VG)

Types of MOSFETS and their modes:

S.No	NMOS	PMOS
01.	Enhancement mode (+VG applied to gate)	Depletion mode (-VG applied to the gate)
02.	Enhancement mode (-VG applied to gate)	Depletion mode (+VG is applied to gate)

Operation Regions in a MOSFET:

Regions	Cutoff region	Triode Region	Saturation Region
Conditions	$(V_{GS} < V_T)$	$(V_{GS} > V_T) \& V_{DS} < V_{GS} - V_T$	$(V_{GS} > V_T) \& V_{DS} \geq V_{GS} - V_T$

Methodology:

Characteristics:

NMOS:

1. Enhancement mode

- Transfer characteristics [ID vs VGS]
 1. Primary Sweep Graph
 2. Parametric graph [ID vs VGS with variable VDS]
- Drain characteristics [ID vs VDS]
 1. Primary sweep graph

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2. Parametric sweep graph [ID vs VDS with variable VGS]

2. Depletion mode

- Transfer characteristics [ID vs VGS]
- 3. Primary Sweep Graph
- 4. Parametric graph [ID vs VGS with variable VDS]
- Drain characteristics [ID vs VDS]
- 3. Primary sweep graph
- 4. Parametric graph [ID vs VDS with variable VGS]

PMOS:

3. Enhancement mode

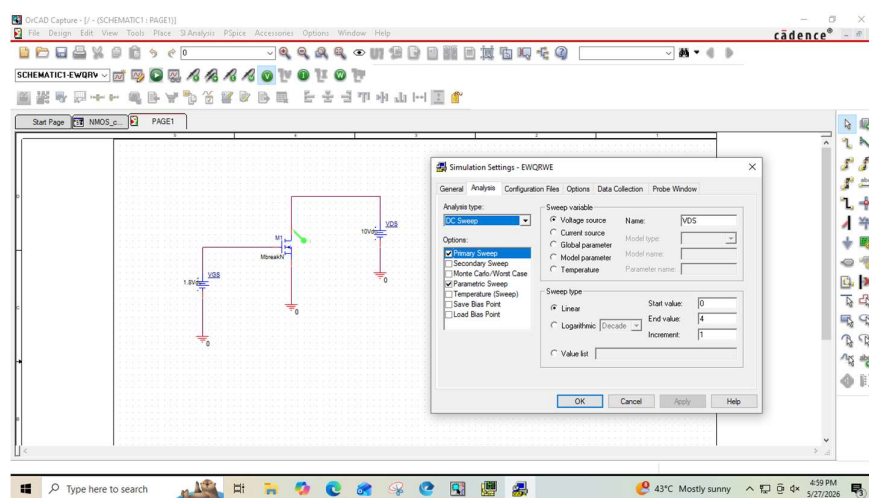
- Transfer characteristics [ID vs VGS]
- 5. Primary Sweep Graph
- 6. Parametric graph [ID vs VGS with variable VDS]
- Drain characteristics [ID vs VDS]
- 5. Primary sweep graph
- 6. Parametric sweep graph [ID vs VDS with variable VGS]

4. Depletion mode

- Transfer characteristics [ID vs VGS]
- 7. Primary Sweep Graph
- 8. Parametric graph [ID vs VGS with variable VDS]
- Drain characteristics [ID vs VDS]
- 7. Primary sweep graph
- 8. Parametric graph [ID vs VDS with variable VGS]

Circuit Diagram in PSpice:

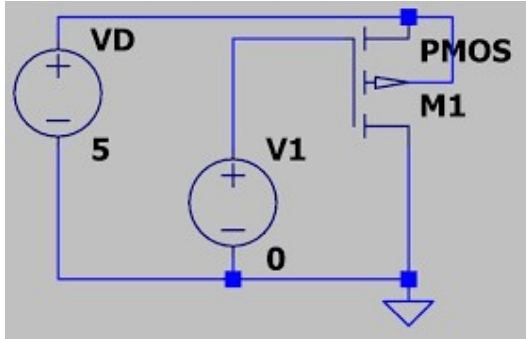
NMOS:



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PMOS:

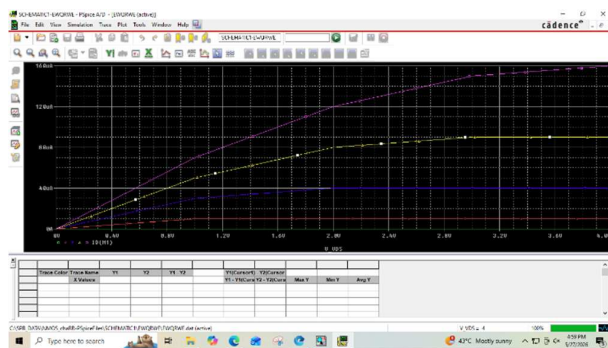


Results:

Graph 01 :

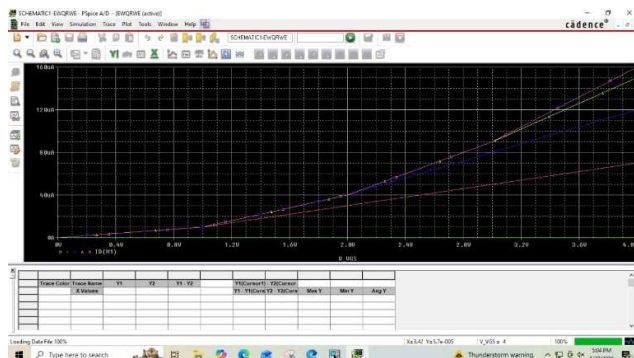
NMOS - Enhancement –

Transfer characteristics (I_D vs V_{GS})



Graph 02 :

NMOS - Enhancement - Drain characteristics (I_D vs V_{DS})



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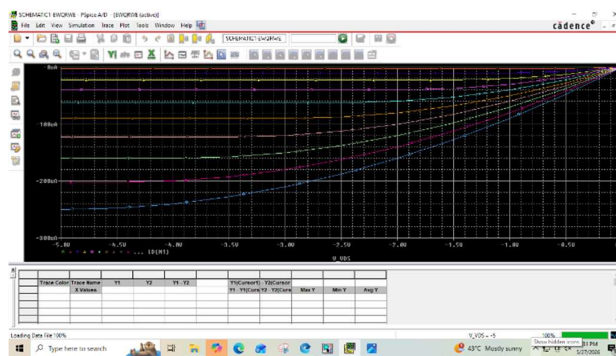
Graph 03 :

PMOS - Enhancement - Transfer characteristics (I_D vs V_{GS})



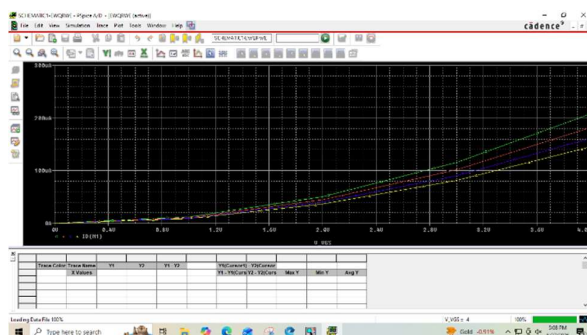
Graph 04 :

PMOS - Enhancement - Drain characteristics (I_D vs V_{DS})



Graph 05 :

Effect of Temperature – NMOS

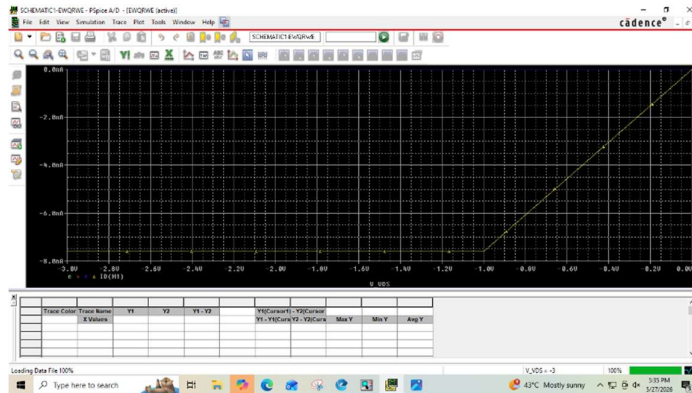


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Graph 06 :

Effect of Temperature – PMOS



DAY-02 [28052026]

Objective:

Implementation of transfer and drain characteristics of PMOS and NMOS in Cadence software, And Implementation of steady and transient state analysis of CMOS

Characteristics:

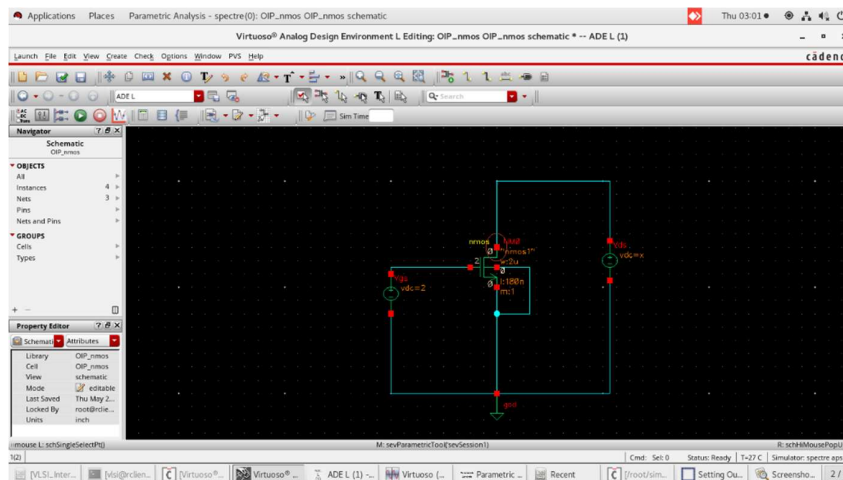
1. NMOS
 - Transfer characteristics (ID vs VGS)
 - Drain Characteristics (ID vs VDS)
2. PMOS
 - Transfer characteristics (ID vs VGS)
 - Drain Characteristics (ID vs VDS)
3. CMOS
 - Dc state analysis
 - Transient state analysis

Circuit Diagrams:

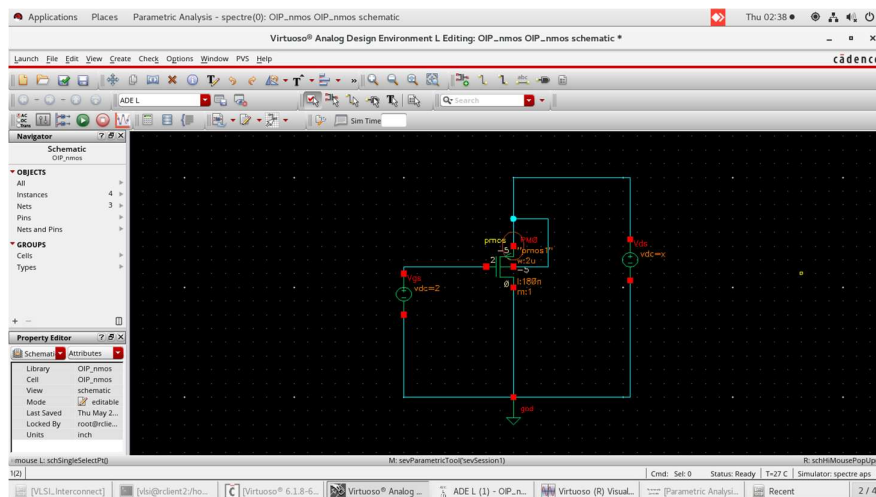
1. NMOS

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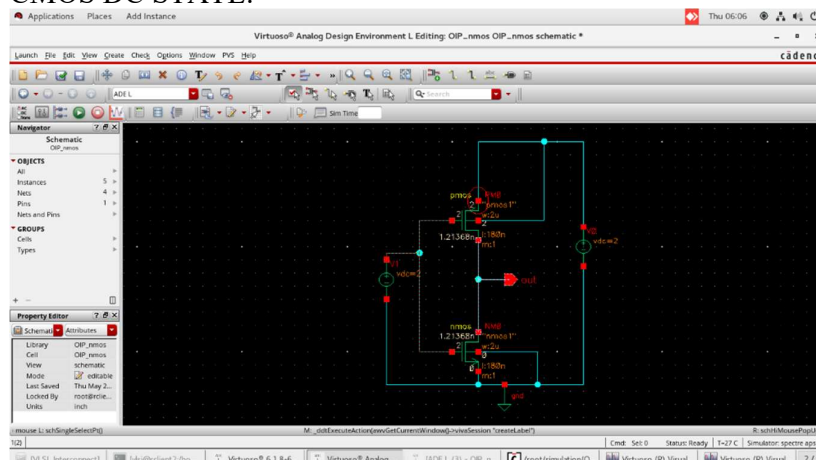
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2. PMOS



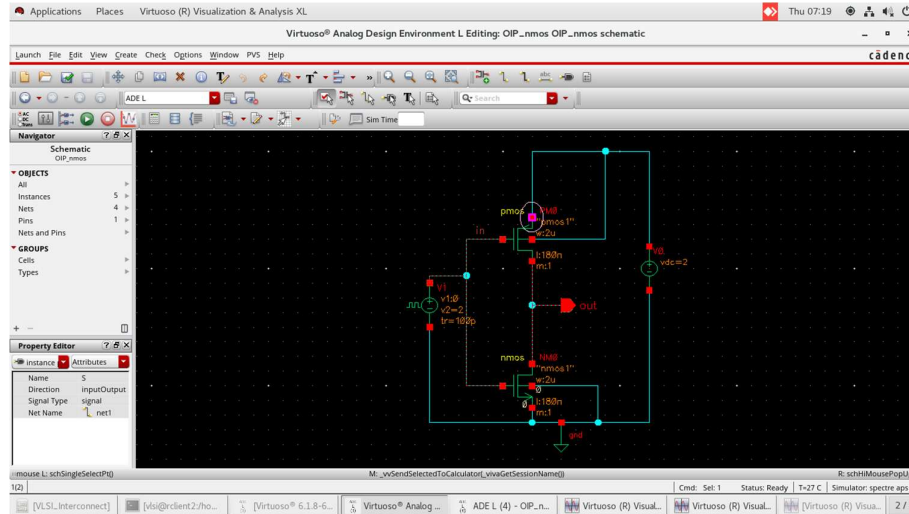
3. CMOS DC STATE:



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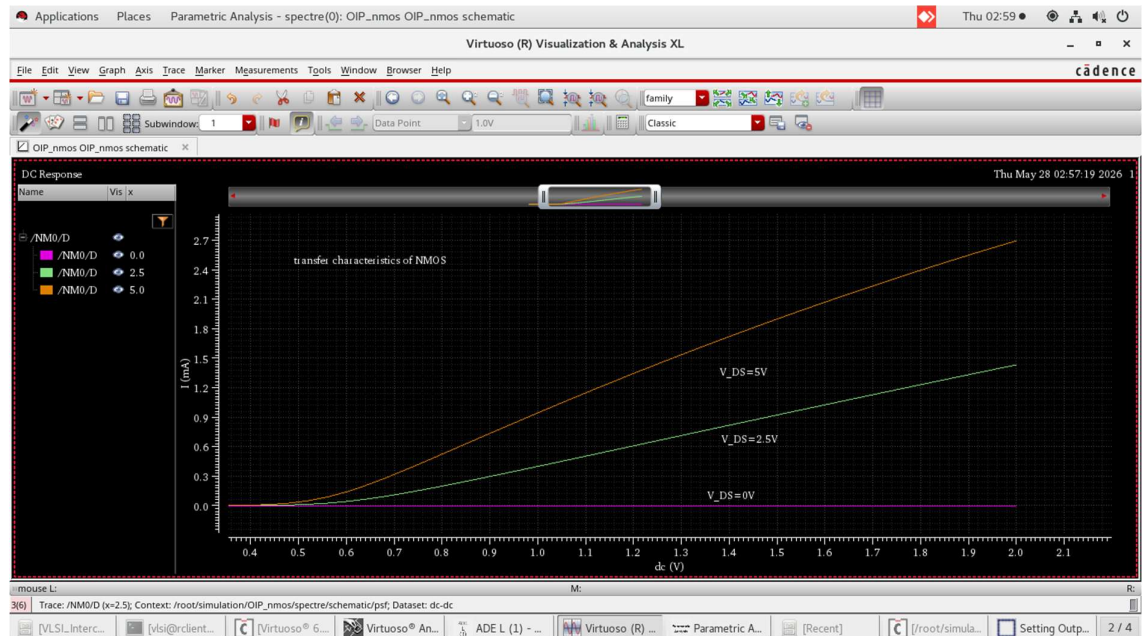
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4. CMOS TRANSIENT STATE



Characteristics graphs:

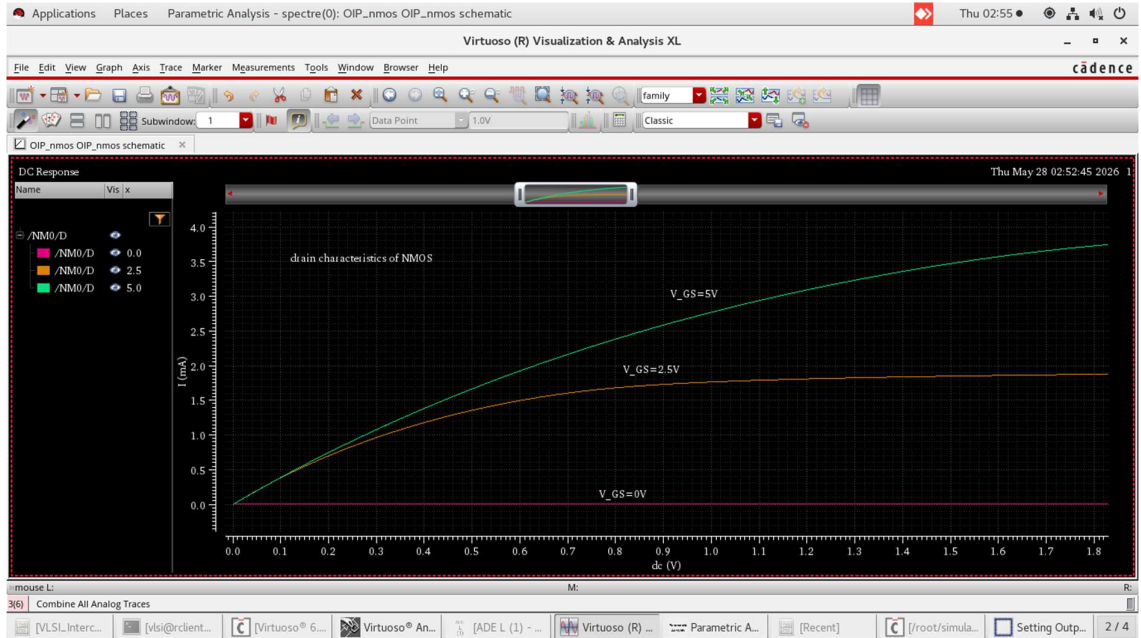
1. NMOS TRANSFER CHARACTERISTICS



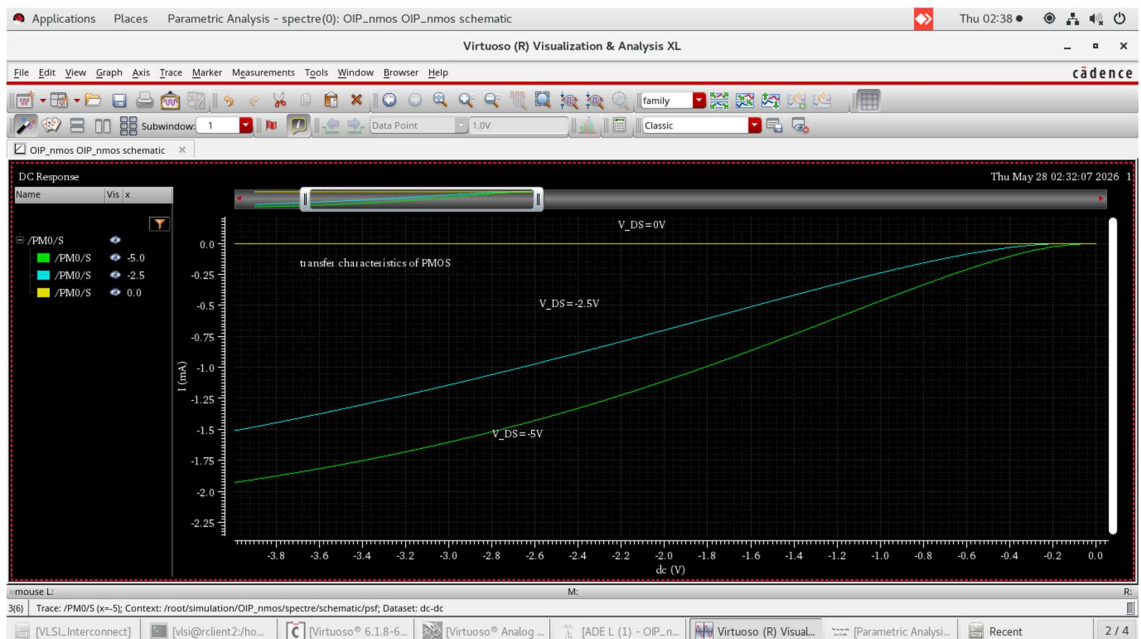
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2. NMOS DRAIN CHARACTERISTICS



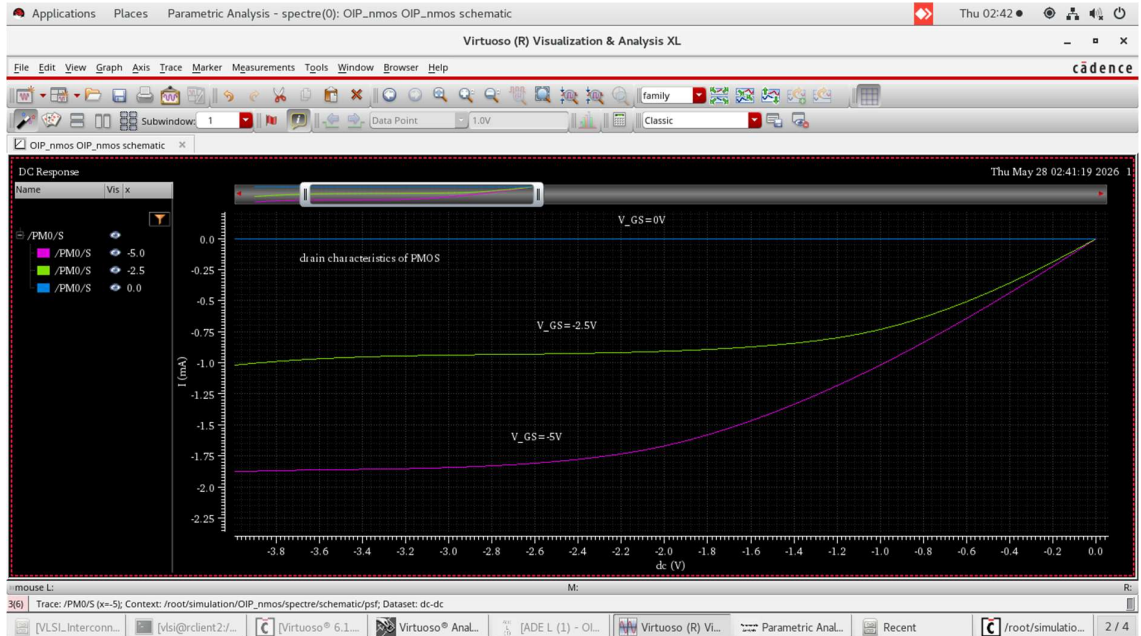
3. PMOS TRANSFER CHARACTERISTICS



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4. PMOS DRAIN CHARACTERISTICS



5. CMOS DC STATE ANALYSIS



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6. CMOS TRANSIENT STATE ANALYSIS

